

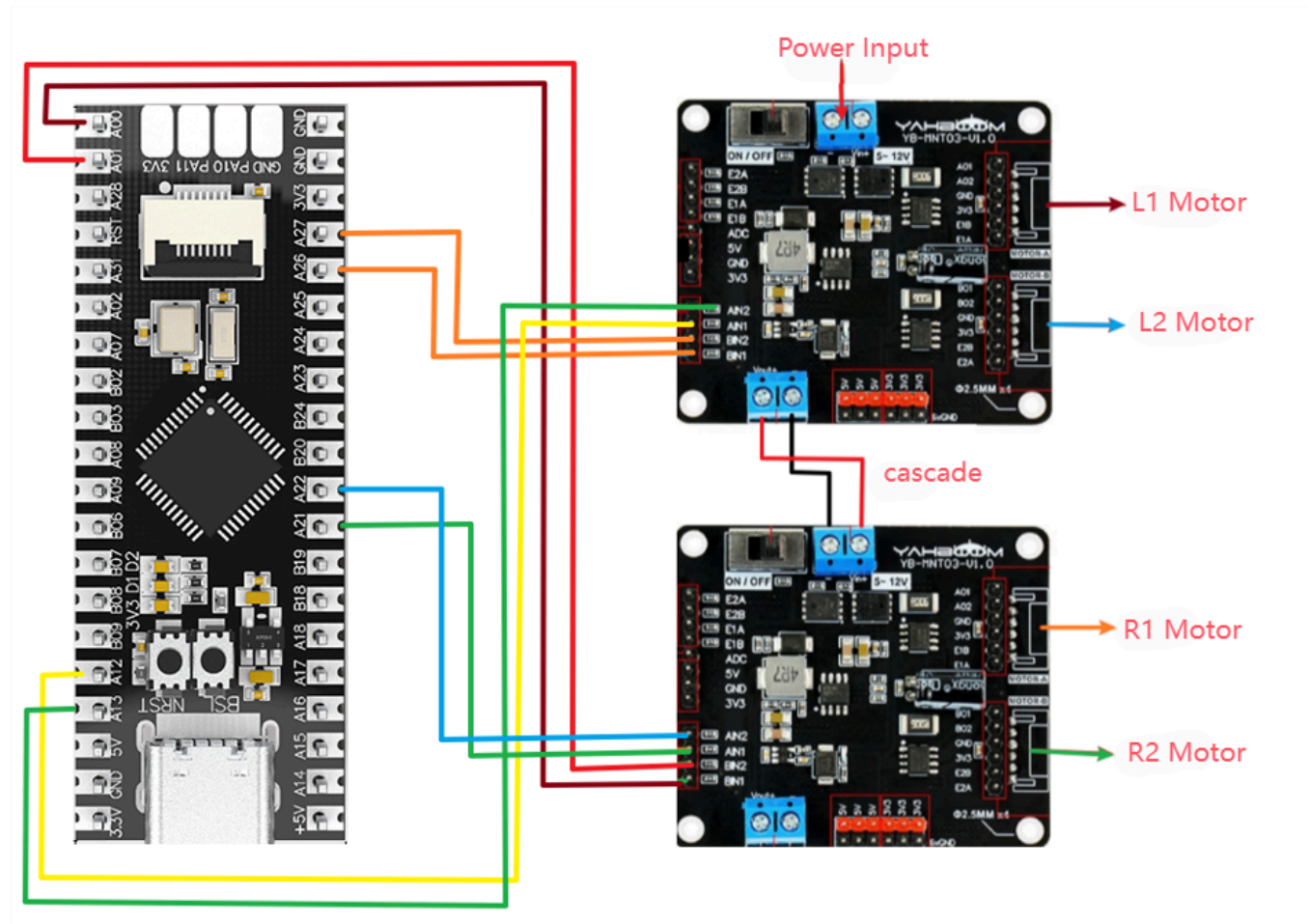
AT8236 2-Channel Motor Driver Module

I. Learning Objectives

Use AT8236 motor driver module to drive the motor.

II. Hardware connection

Pin connection between MSPM0G3519 and AT8236



L1 motor:

MSPM0G3519	AT8236
PA12	AIN1
PA13	AIN2

L2 motor:

MSPM0G3519	AT8236
PA26	BIN1
PA27	BIN2

R1 motor:

MSPM0G3519	AT8236
PA21	AIN1
PA22	AIN2

R2 motor:

MSPM0G3519	AT8236
PA0	BIN1
PA1	BIN2

Motor and voltage regulator board pin connection:

AT8236	Motor
AO1	Motor power +
AO2	Motor power -

AT8236 motor driver module voltage regulator board pin description:

Pin details					
Interface type	Pin name	Pin description	Interface type	Pin name	Pin description
MCU/ host interface	E1A	Motor 1 Hall signal A	Motor port	AO1	Motor 1 power supply+
	E1B	Motor 1 Hall signal B		AO2	Motor 1 power supply-
	E2A	Motor 2 Hall signal A		GND	GND
	E2B	Motor 2 Hall signal B		3V3	Motor 1 Hall power supply
	ADC	Collect VM input voltage		E1B	Motor 1 Hall signal B
	5V	Output 5V3A power supply		E1A	Motor 1 Hall signal A
	GND	GND		BO1	Motor 2 power supply+
	3V3	Output 3.3V voltage		BO2	Motor 2 power supply-
	AIN1	Motor 1 drive signal 1		GND	GND
	AIN2	Motor 1 drive signal 2		3V3	Motor 2 Hall power supply
	BIN1	Motor 2 drive signal 1		E2B	Motor 2 Hall signal B
	BIN2	Motor 2 drive signal 2		E2A	Motor 2 Hall signal A

III. Program description

- bsp_at8236.h

```

#ifndef __BSP_TB6612_H_
#define __BSP_TB6612_H_

#include "ti_msp_dl_config.h"

void init_motor(void);

void L1_control(uint16_t motor_speed,uint8_t dir);
void L2_control(uint16_t motor_speed,uint8_t dir);
void R1_control(uint16_t motor_speed,uint8_t dir);
void R2_control(uint16_t motor_speed,uint8_t dir);

#endif

```

Define four motor control functions.

- bsp_at8236.c

```

void L1_control(uint16_t motor_speed,uint8_t dir)
{
    if(dir)
    {
        DL_TimerA_setCaptureCompareValue(PWM_L1_INST, motor_speed, DL_TIMER_CC_0_INDEX);
        DL_TimerA_setCaptureCompareValue(PWM_L1_INST, 0, DL_TIMER_CC_1_INDEX);
    }
    else
    {
        DL_TimerA_setCaptureCompareValue(PWM_L1_INST, 0, DL_TIMER_CC_0_INDEX);
        DL_TimerA_setCaptureCompareValue(PWM_L1_INST, motor_speed, DL_TIMER_CC_1_INDEX);
    }
}

```

The L1_control function is used to control the speed and direction of the L1 motor by adjusting the duty cycle of the PWM signal.

motor_speed and dir represent the motor speed and motor direction of the motor respectively.

Note: The project source code must be placed in the SDK path for compilation,

For example, the path: D:\TI\M0_SDK\mspm0_sdk_1_30_00_03\TB6612

> 新加卷 (D:) > TI > M0_SDK > msp0_sdk_1_30_00_03				
名称	修改日期	类型	大小	
1.TB6612	2024/7/22 18:59	文件夹		
2.AT8236	2024/7/22 19:47	文件夹		
3.Enconder	2024/7/23 10:36	文件夹		
4.Servo	2024/7/23 11:13	文件夹		
docs	2024/7/23 10:33	文件夹		
examples	2024/7/23 10:34	文件夹		
kernel	2024/7/23 10:37	文件夹		
source	2024/7/23 10:33	文件夹		
tools	2024/7/23 10:33	文件夹		
imports.mak	2024/1/25 11:45	MAK 文件	2 KB	
known_issues_FAQ.html	2024/1/25 11:42	Microsoft Edge ...	67 KB	
license_msp0_sdk_1_30_00_03.txt	2024/1/25 11:42	文本文档	33 KB	
manifest_msp0_sdk_1_30_00_03.html	2024/1/25 11:42	Microsoft Edge ...	113 KB	
mspm0sdk_1_30_00_03.log	2024/7/23 10:42	文本文档	5,237 KB	
release_notes_msp0_sdk_1_30_00_0...	2024/1/25 11:42	Microsoft Edge ...	108 KB	
uninstall.dat	2024/7/23 10:39	DAT 文件	344 KB	
uninstall.exe	2024/7/23 10:39	应用程序	6,048 KB	